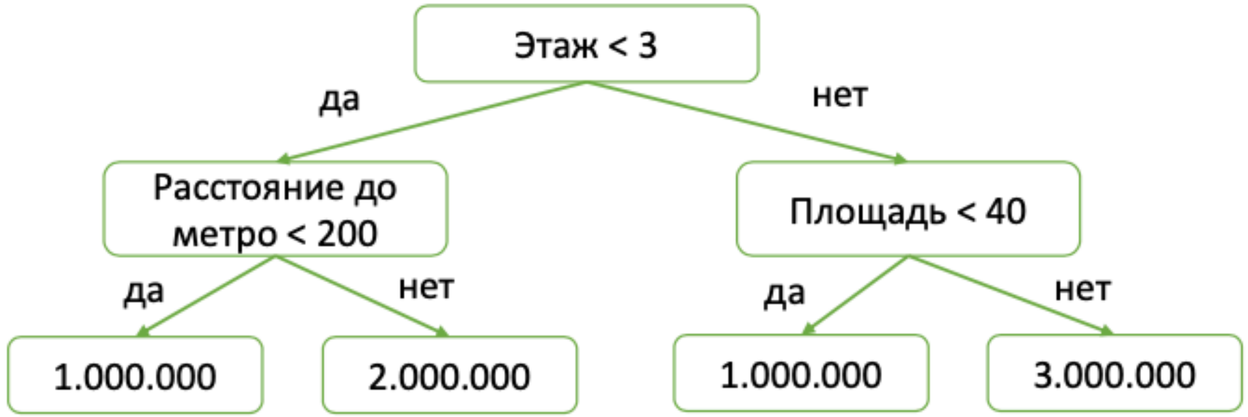
Занятие IX

Decision Trees

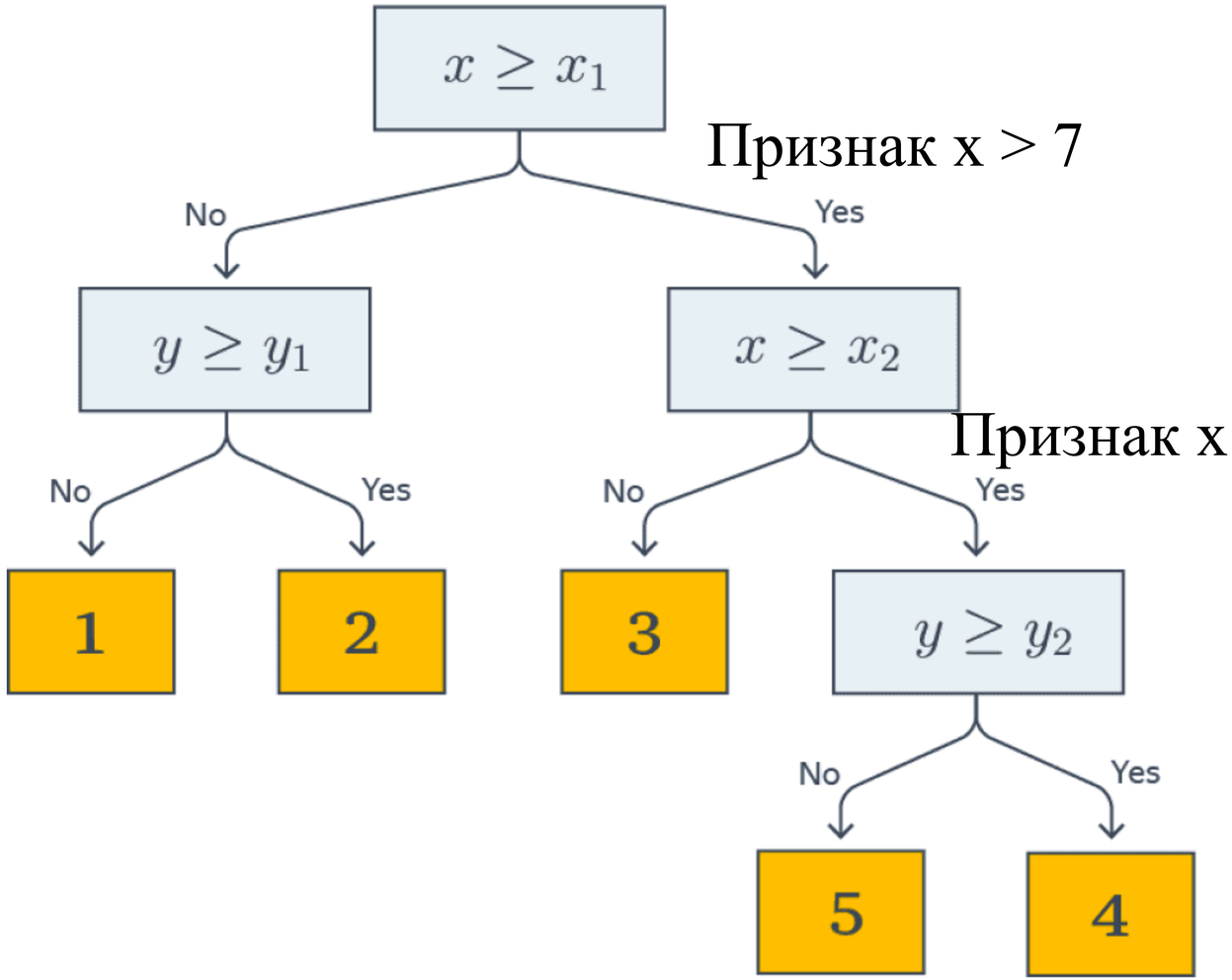
1. **Key** points

- $[30 < \text{площадь} < 50][2 < \text{этаж} < 5][500 < \text{расстояние до метро} < 1000]$
- Легко объяснить, как работают
- Находят нелинейные закономерности



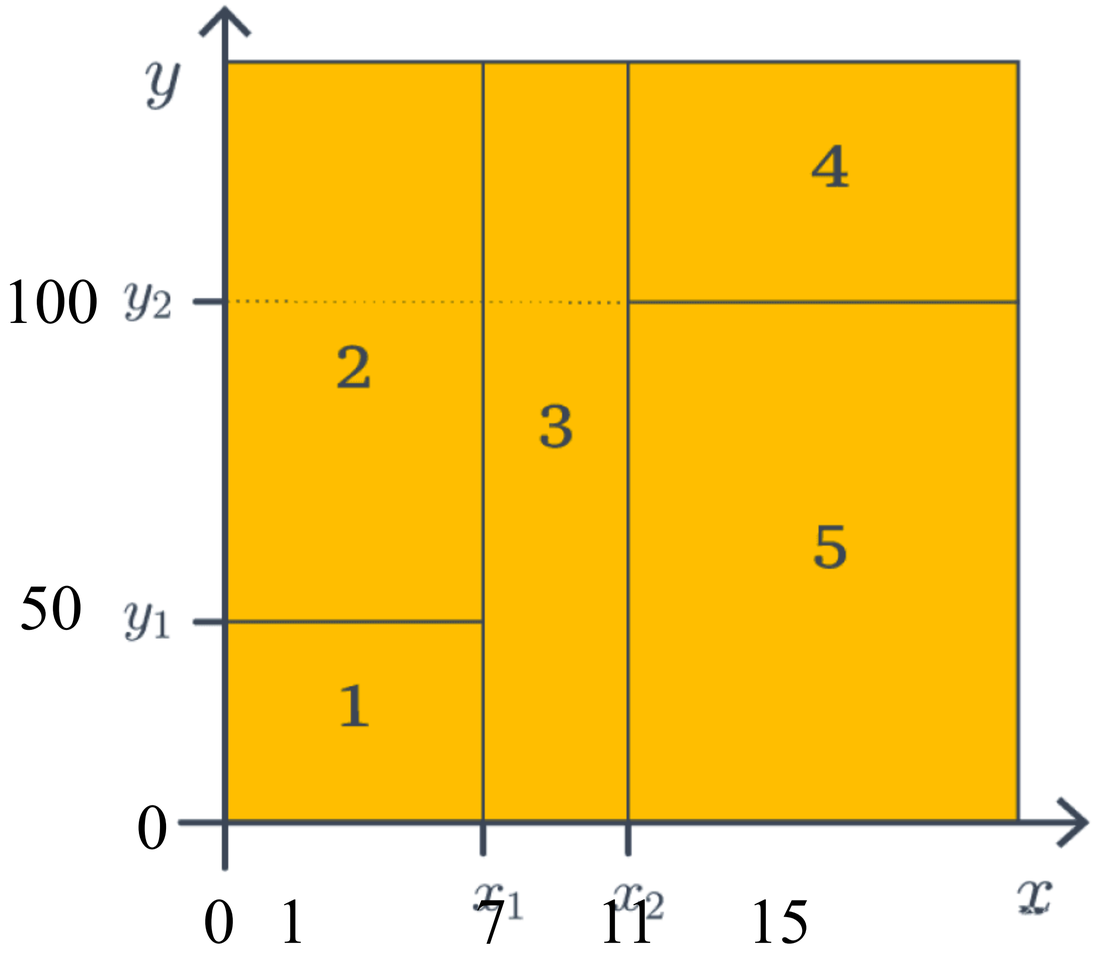
- Внутренние вершины: предикаты $[x_j < t]$
- Листья: прогнозы $c \in \mathbb{Y}$

1. **Key** points



Признак x > 7

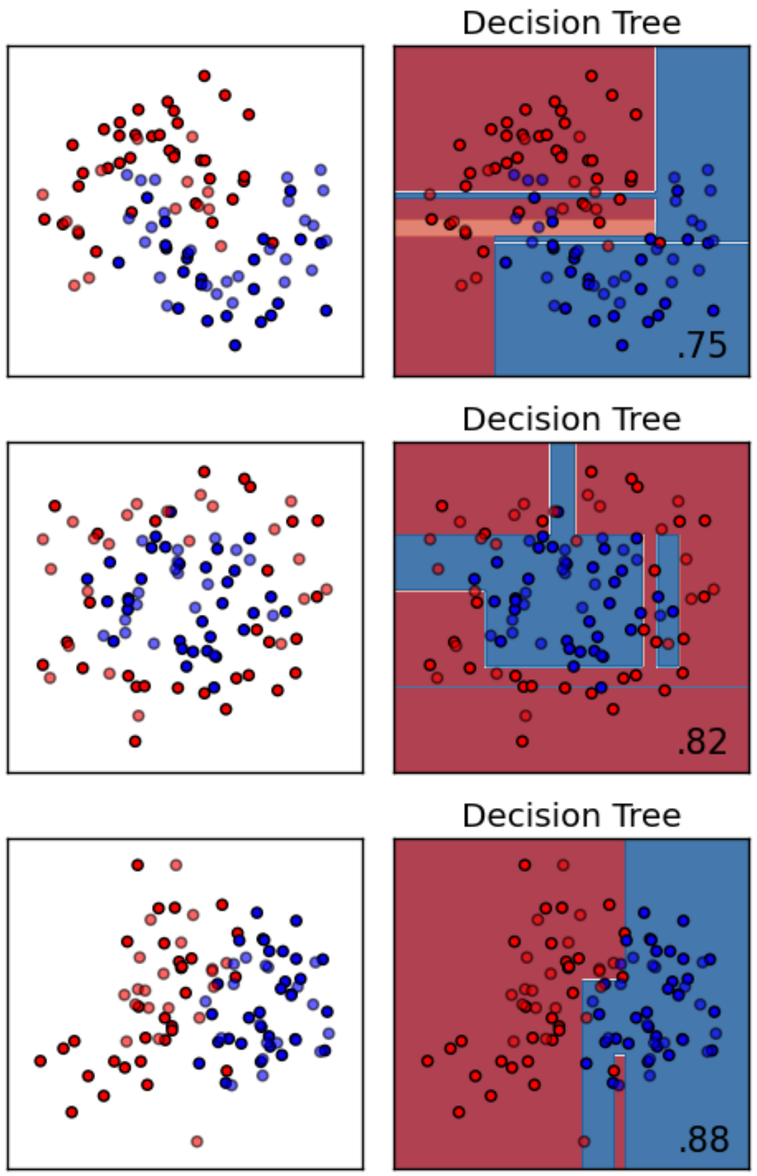
Признак x > 11



1. **Key** points

- *Binary Tree with nodes*
  - *Internal:* $\beta_v : \mathbb{X} \to \{0,1\}$
  - *Leaf:* *prediction* $c_v \in \mathbb{Y}$
    - *constant value for regression*
    - *constant class for classification*
    - *probability vector*

- *Predict to choose direction of split according to predicate:*
  - *Basic:* $\beta_v(x) = [x_j < t]$
  - *Flex:* $\beta_v(x) = [\langle w_v, x \rangle < t]$
  - *Dist:* $\beta_v(x) = [\rho(x, x_v) < t]$



1. Key points

Можно даже линейную модель!

- Наш выбор: константные прогнозы $c_v \in \mathbb{Y}$

- Регрессия:

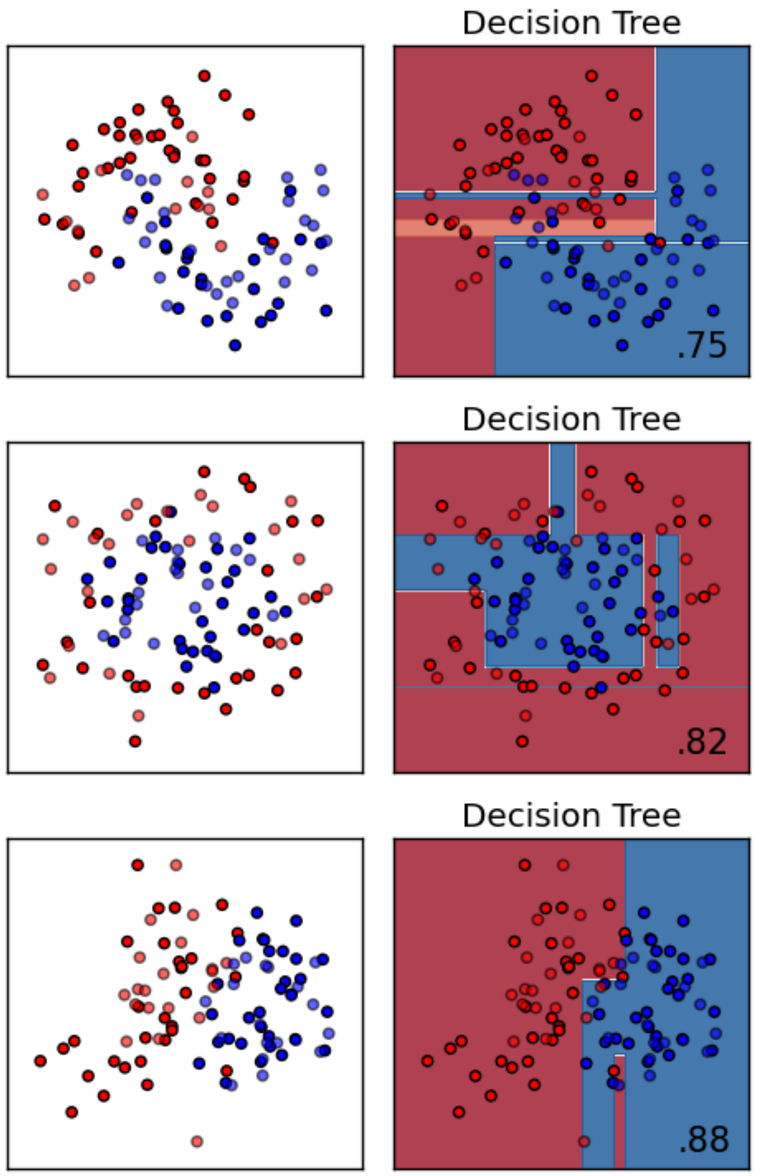
$$c_v = \frac{1}{|R_v|} \sum_{(x_i, y_i) \in R_v} y_i$$

- Классификация:

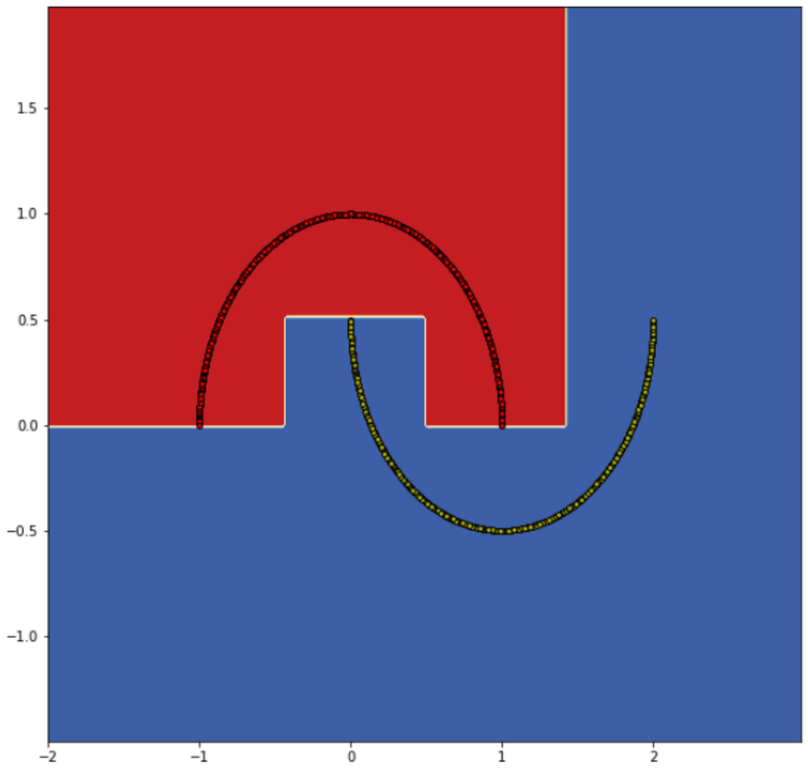
$$c_v = \arg\max_{k \in \mathbb{Y}} \sum_{(x_i, y_i) \in R_v} [y_i = k]$$

- Классификация и вероятности классов:

$$c_{vk} = \frac{1}{|R_v|} \sum_{(x_i, y_i) \in R_v} [y_i = k]$$



1. **Key** points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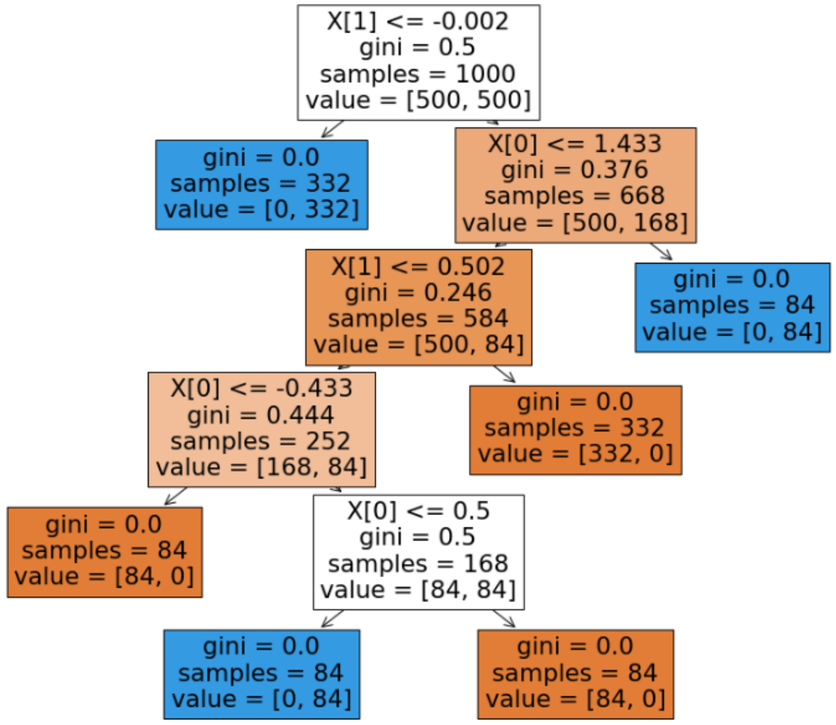
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gini = 0.5
samples = 1000
value = [500, 500]
gini = 0.0
samples = 332
value = [0, 332]
X[0] <= 1.433
gini = 0.376
samples = 668
value = [500, 168]
X[1] <= 0.502
gini = 0.246
samples = 584
value = [500, 84]
gini = 0.0
samples = 84
value = [0, 84]
X[0] <= -0.433
gini = 0.444
samples = 252
value = [168, 84]
gini = 0.0
samples = 332
value = [332, 0]
gini = 0.0
samples = 84
value = [84, 0]
X[0] <= 0.5
gini = 0.5
samples = 168
value = [84, 84]
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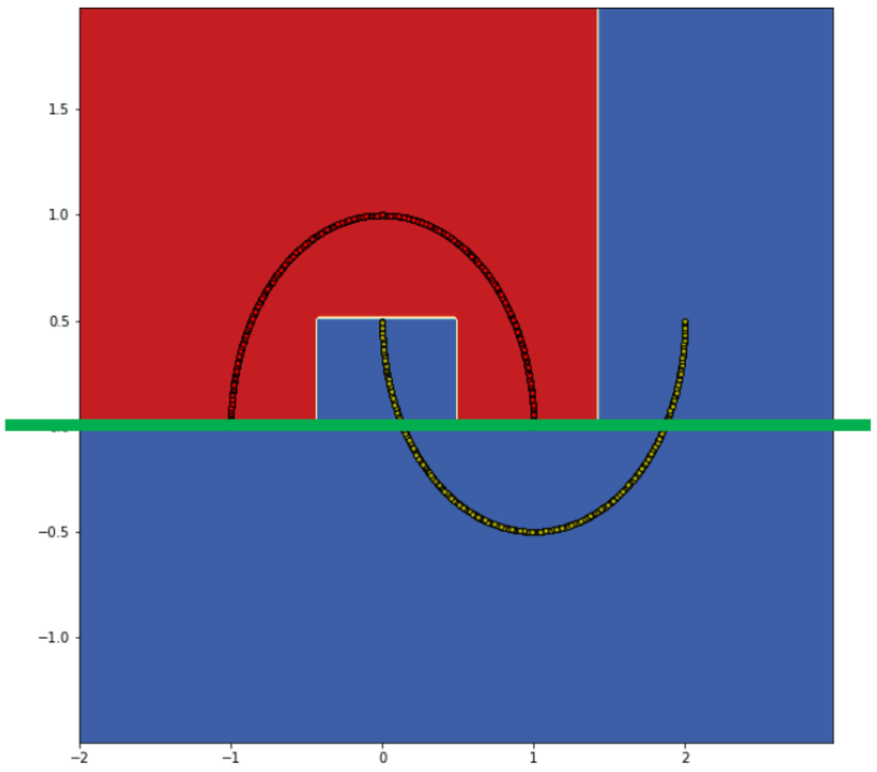
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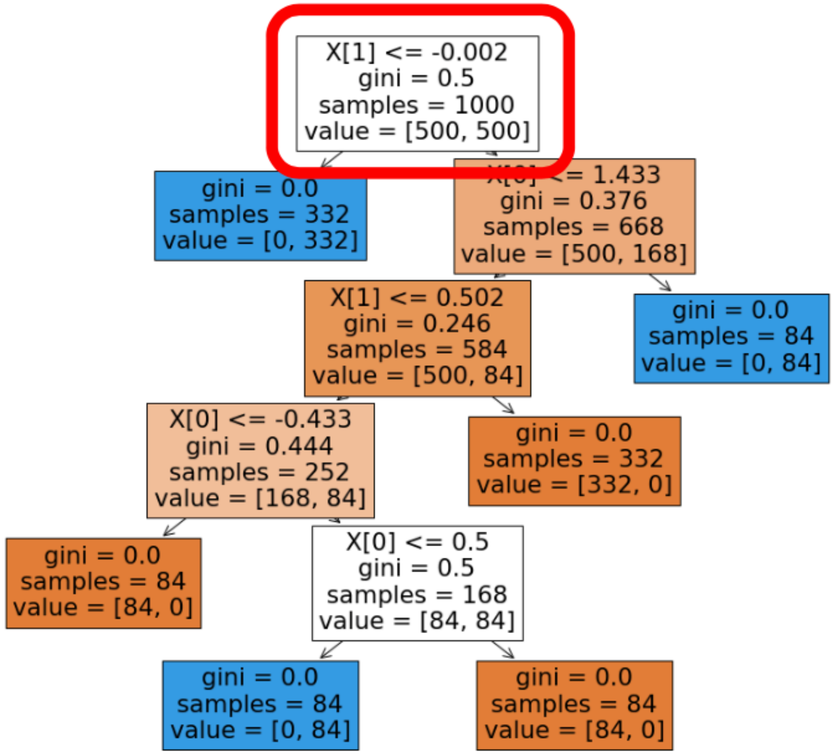
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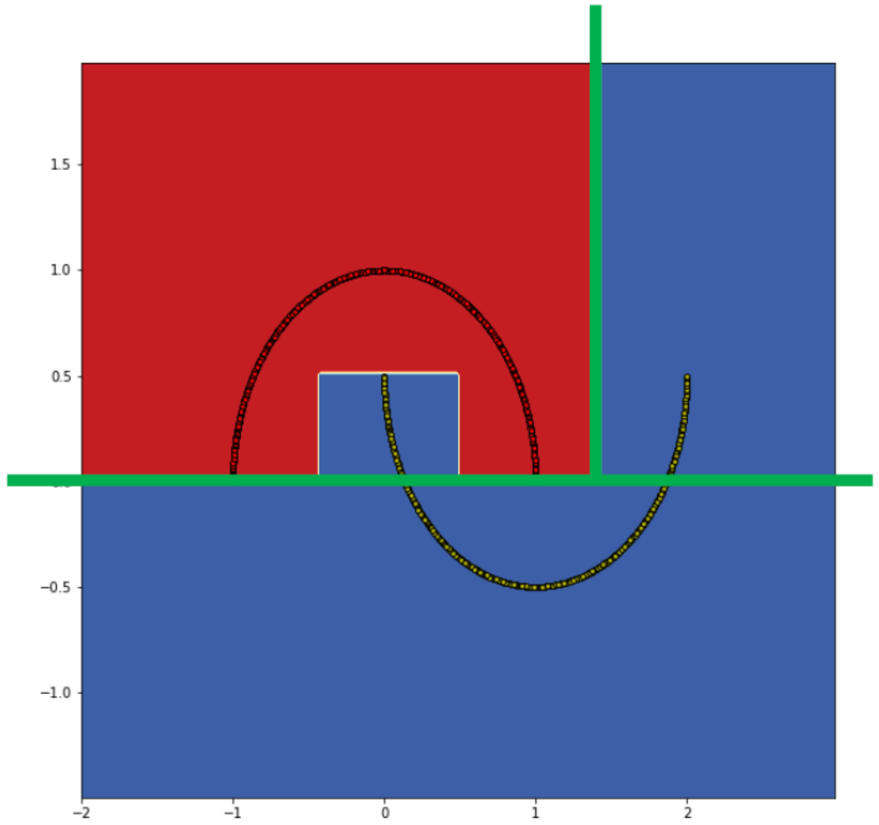
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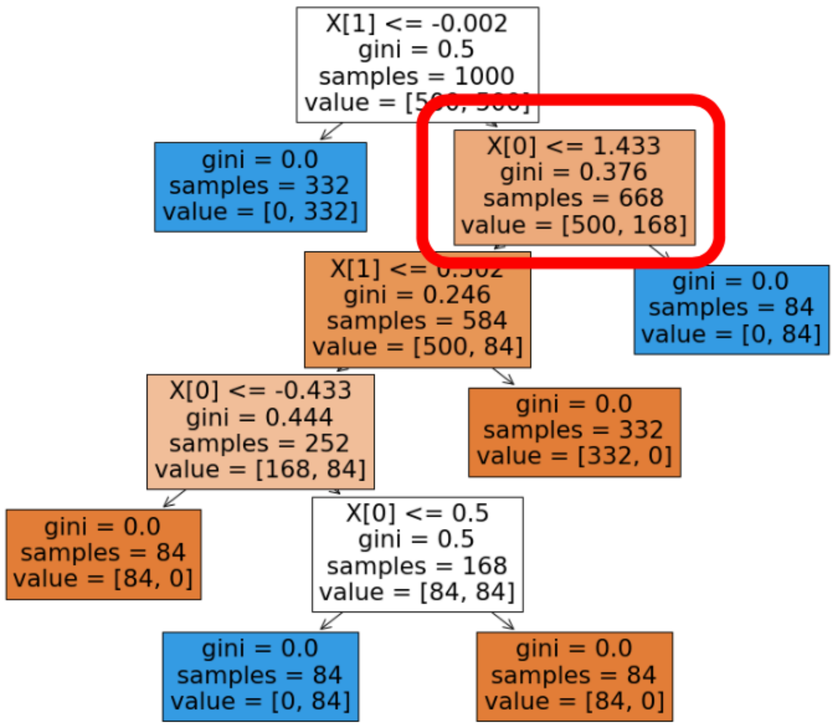
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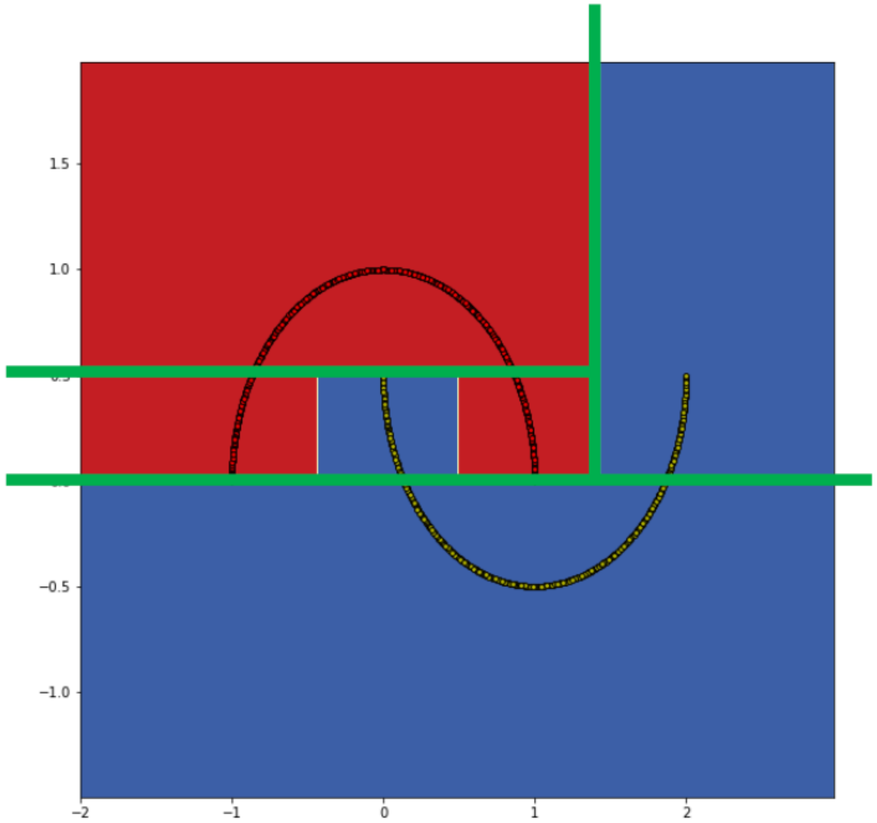
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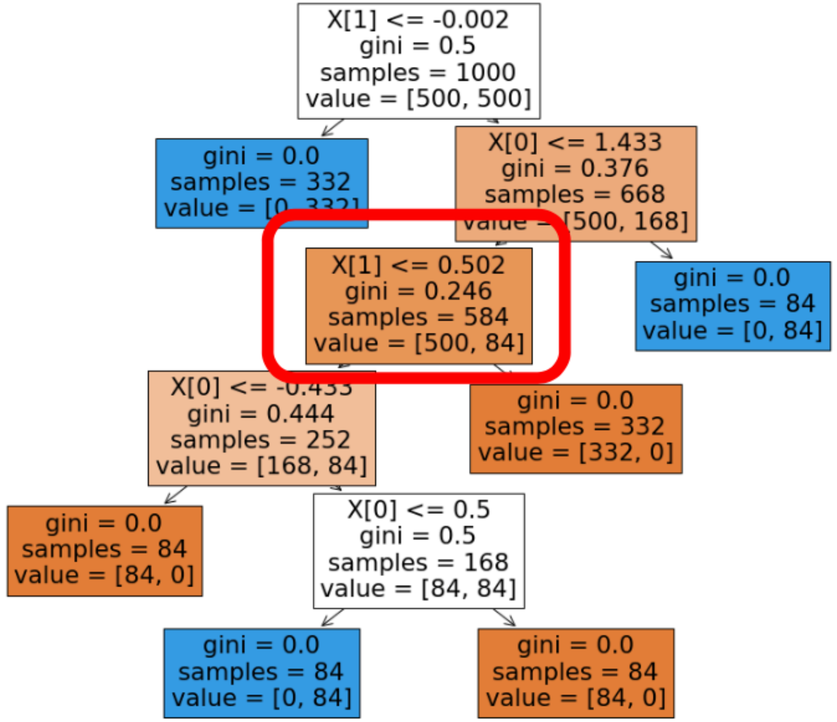
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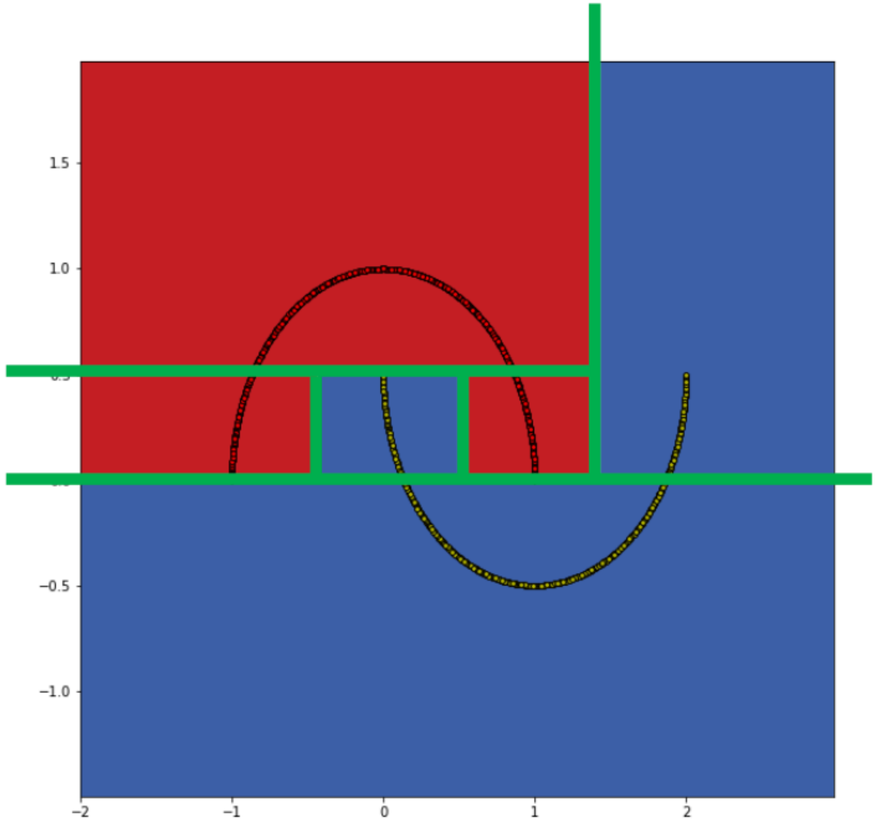
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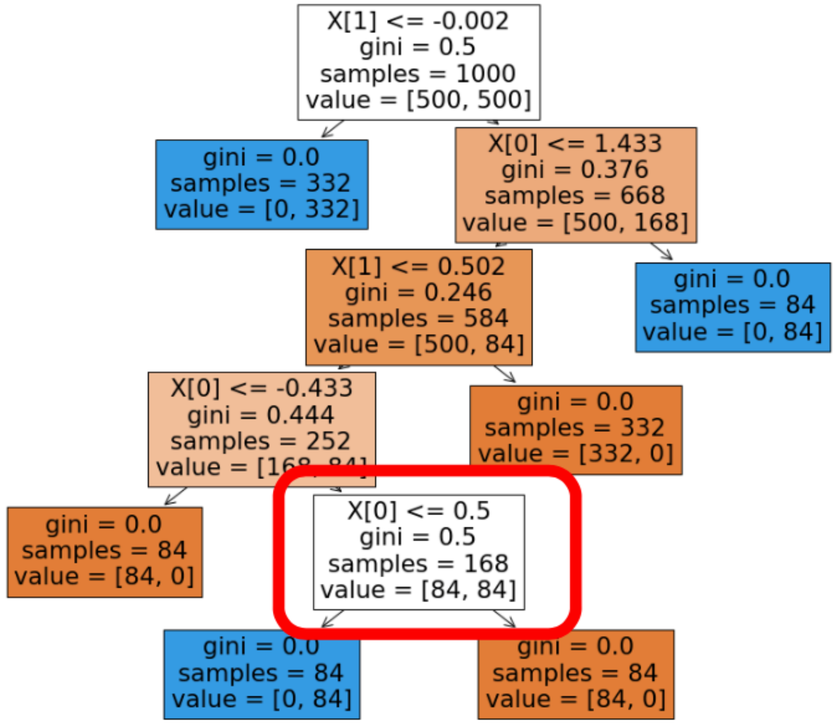
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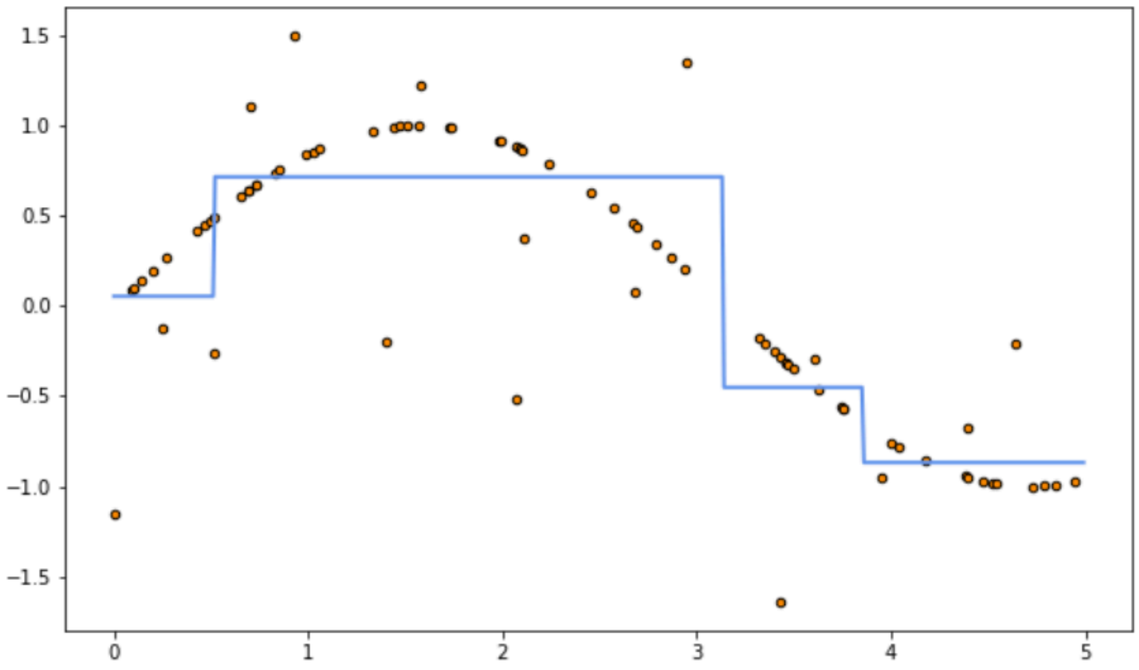
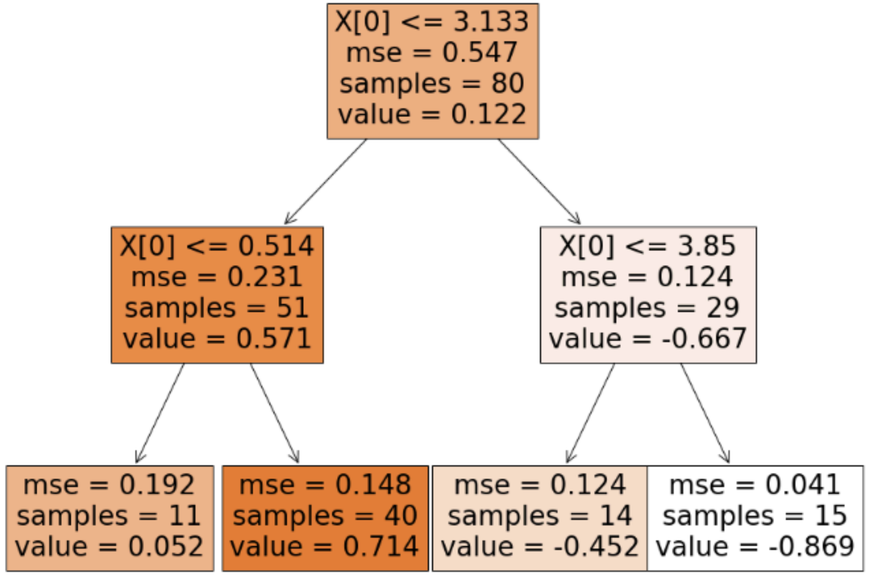
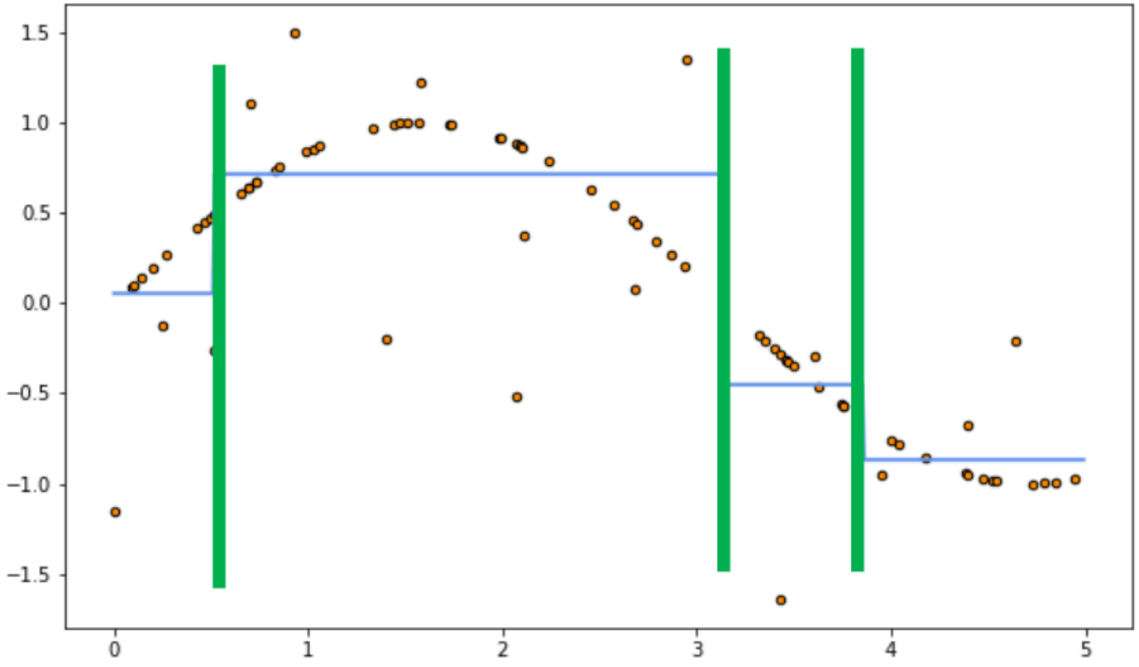
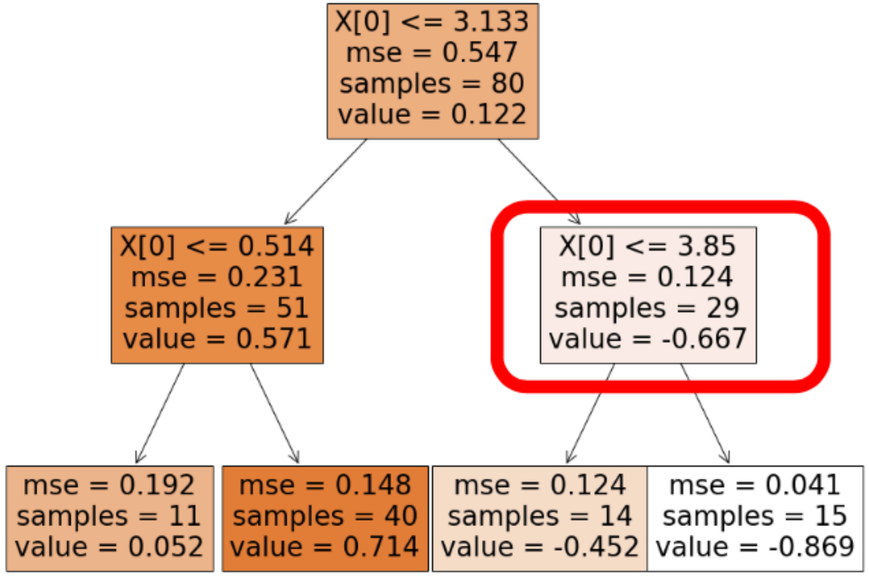
1. **Key** points/regression





1. **Key** points/regression



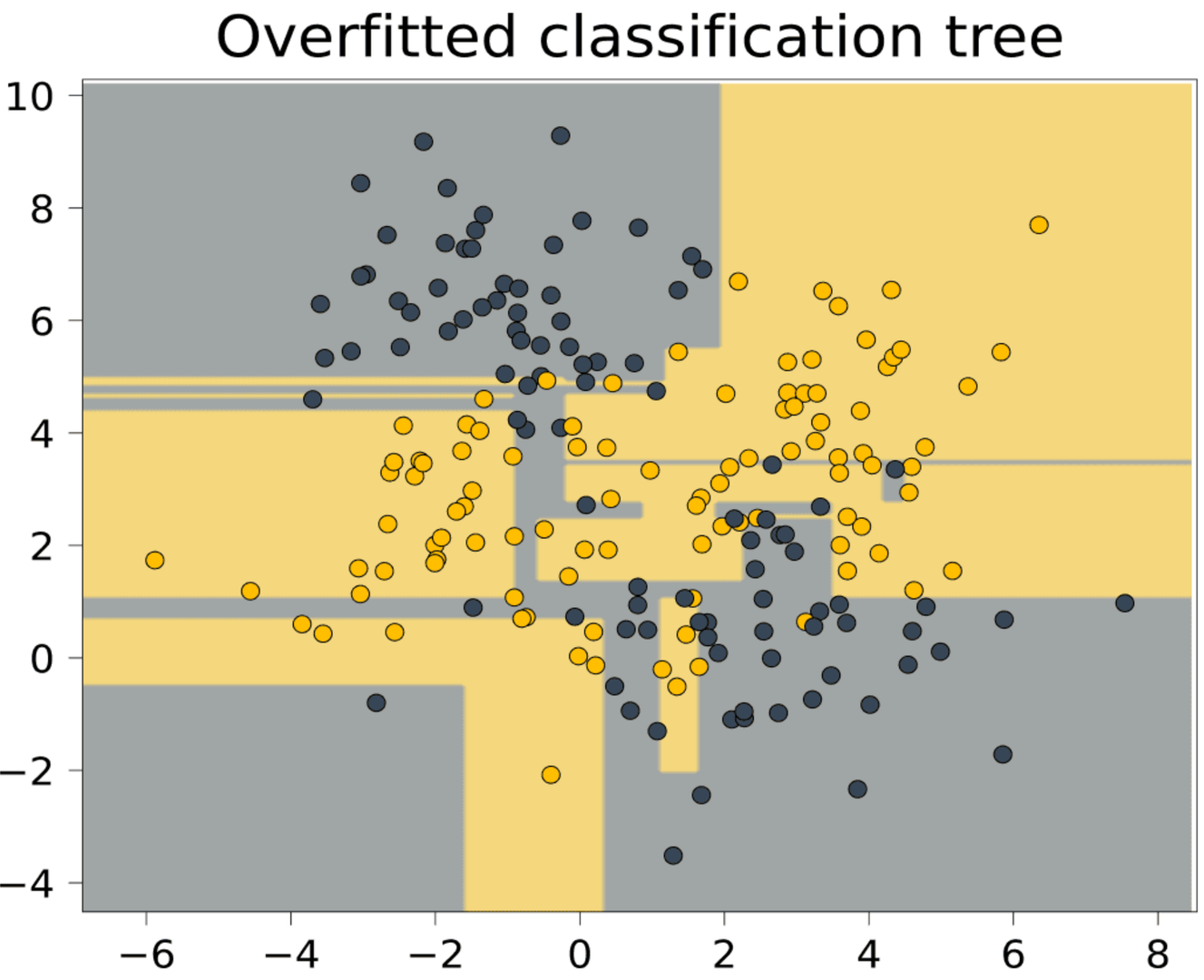


1. **Key** points

- Дерево разбивает признаковое пространство на области $R_1, \dots, R_J$
- Каждая область R_j соответствует листу
- В области R_j прогноз c_j константный

$$a(x) = \sum_{j=1}^{J} c_j \left[x \in R_j\right]$$

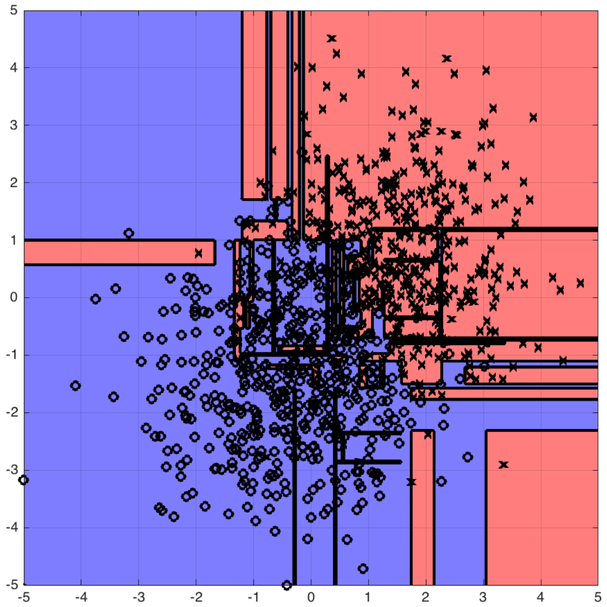
2. **Over**Training

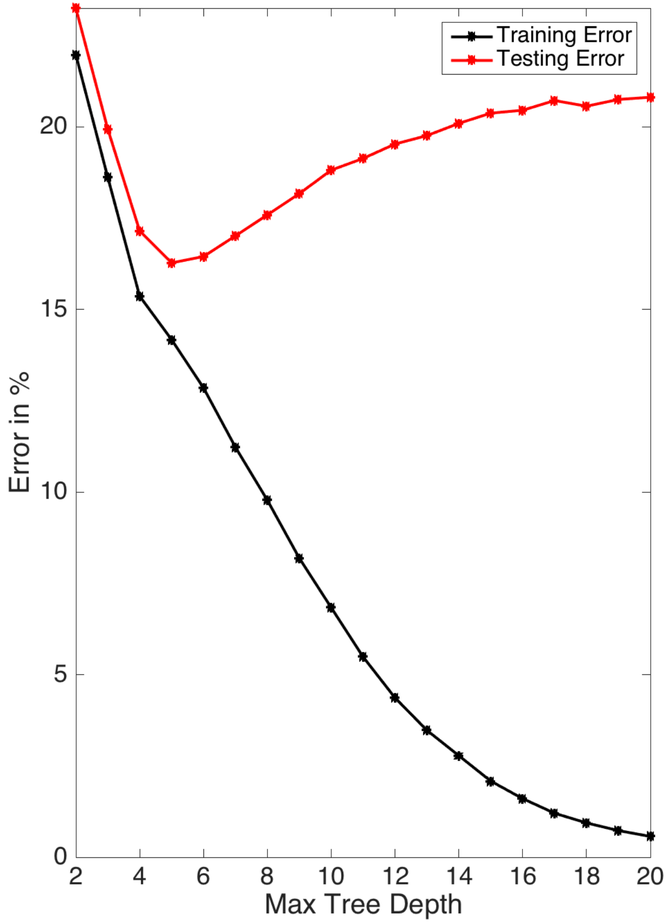


- *We can easily make such a tree that will have perfect quality*
- *Simply put each object into separate leaf*

- *But:*
  - *Such a tree will be overfitted*
  - *It will not be able to work with unseen before data scope*

2. **Over**Training





We can easily make such a tree that will have perfect quality

Simply put each object into separate leaf

But:

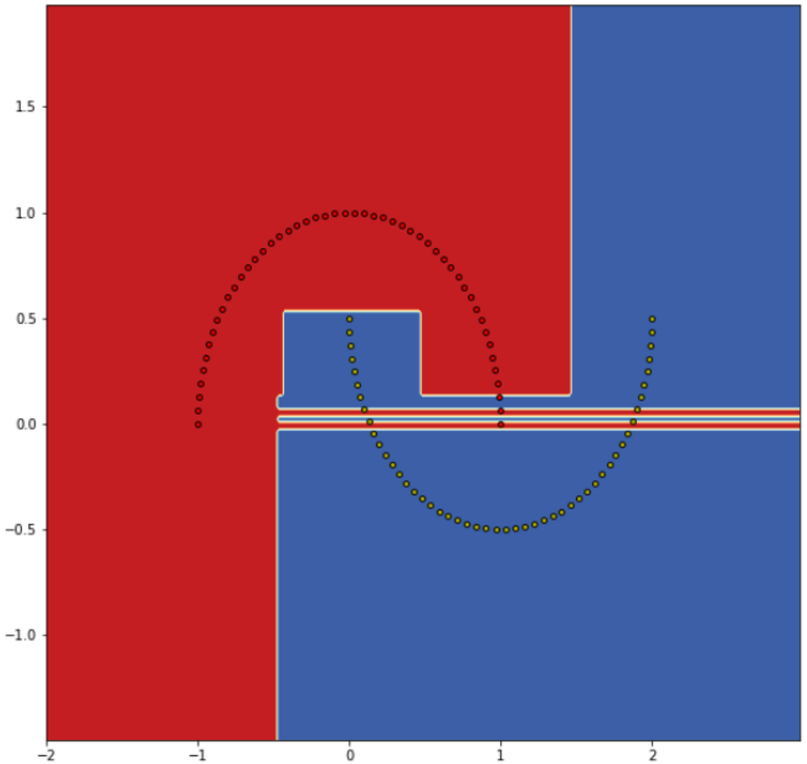
- *Such a tree will be overfitted*
- *It will not be able to work with unseen before data scope*

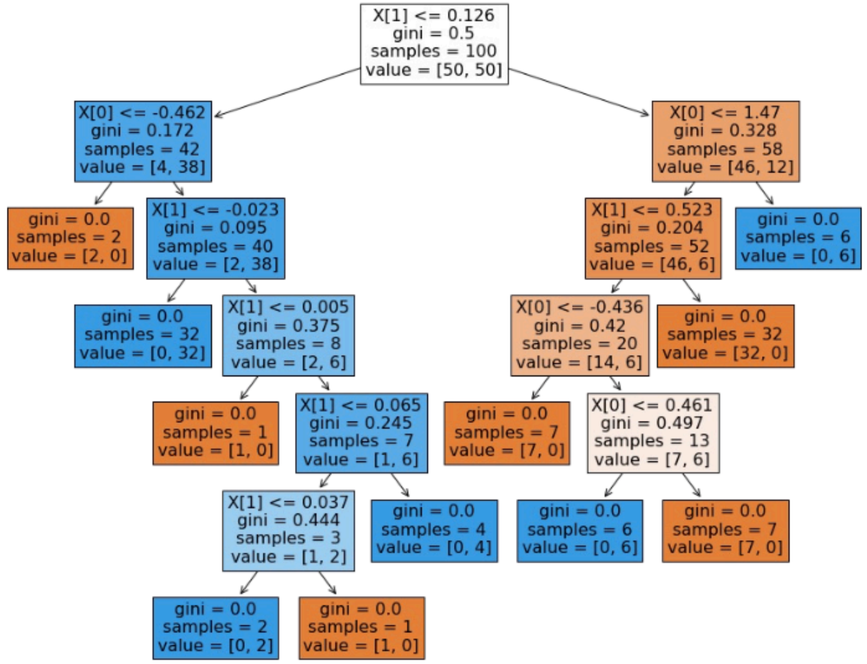
2. **Over**Training

- Решающее дерево можно строить до тех пор, пока каждый лист не будет соответствовать ровно одному объекту

- Деревом можно идеально разделить любую выборку!

- Если только нет объектов с одинаковыми признаками, но разными ответами





2. **Over**Training

- Решающее дерево можно строить до тех пор, пока каждый лист не будет соответствовать ровно одному объекту
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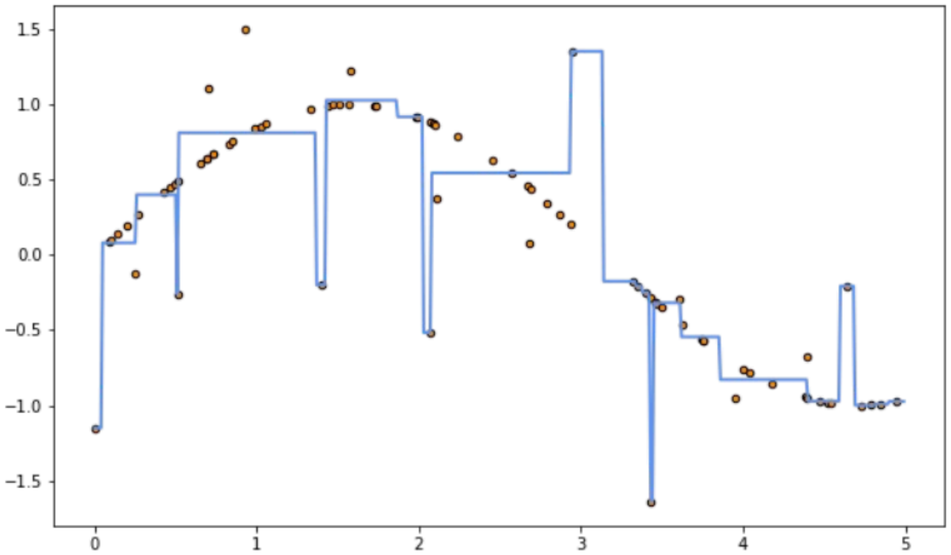
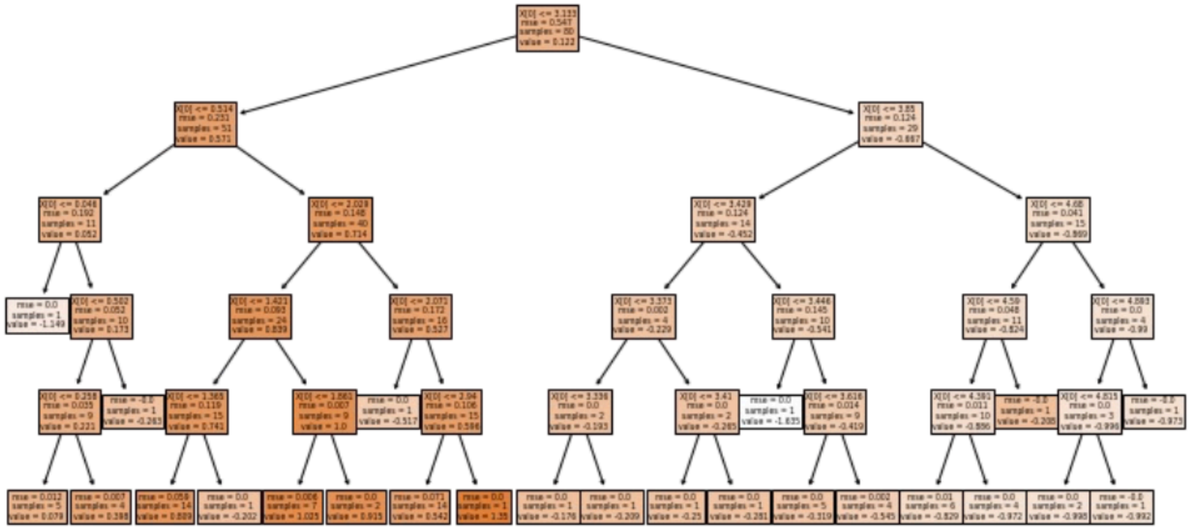
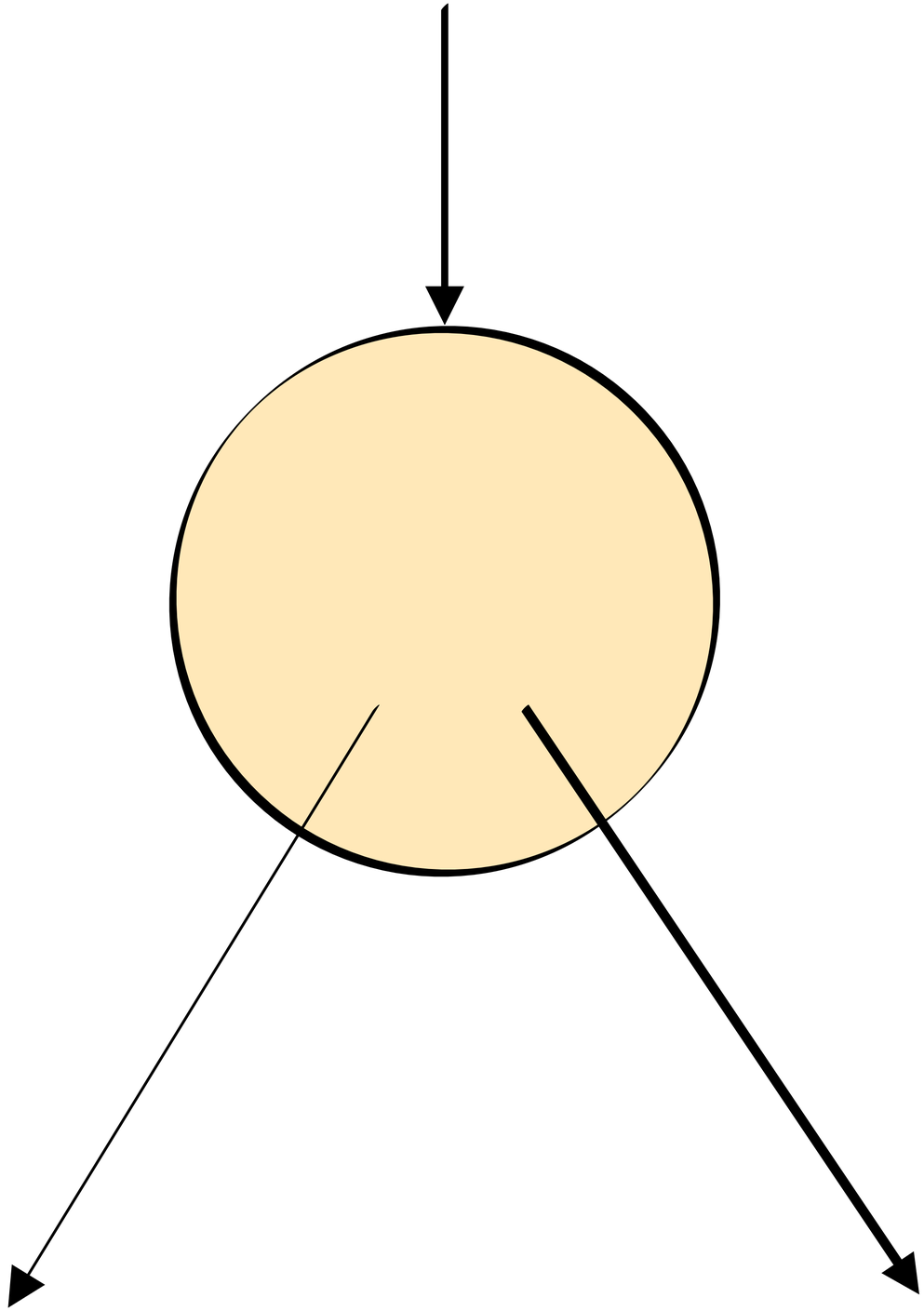




3. **Optimal** Tree

- *No overfitting!*
- *Best quality with minimum height / № of nodes*
    - *To find such - NP-complete problem*
- *Let's construct tree be levels from root with Greedy Approach*
- *Parameters to focus on:*
    - *Height of the tree*
    - *№ of nodes*
    - *Predicates*
    - *Leaves' predictions*



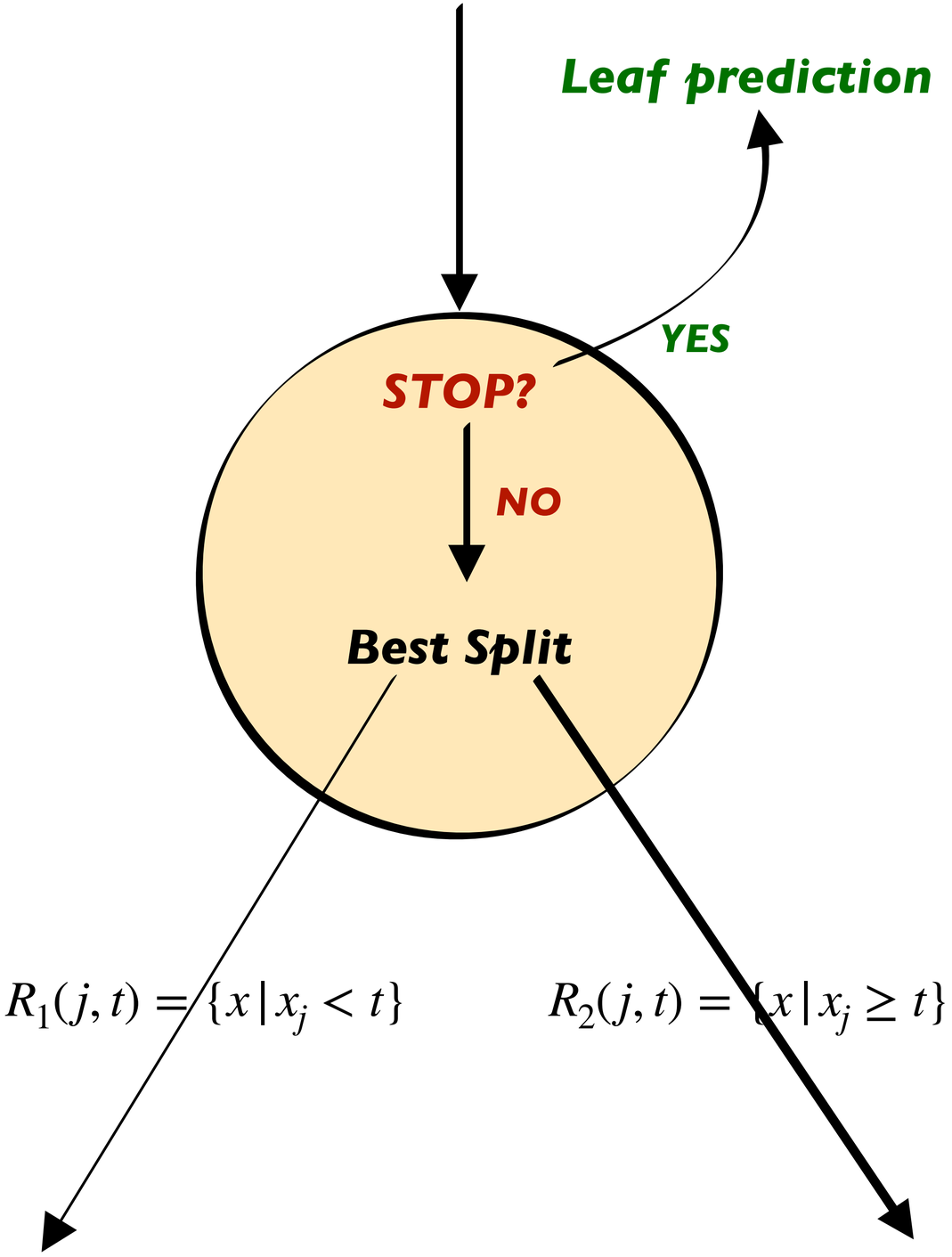
4. **Greedy** building

- *Define Split function to find the best predicate*
    - *We will look for such according to Quality estimate*

- *We can find* $R_1(j,t) = \{x \mid x_j < t\}$ ***&*** $R_2(j,t) = \{x \mid x_j \geq t\}$ ***- left & right subtrees***

- *Repeat operation: find optimal split — split data into two subtrees*

- *We need to check if the node should be the* ***Leaf - Stop function***



4. **Greedy** building

FitNode (R_m, m):

If Stop_Criteria:

C_m - prediction

m - leaf node

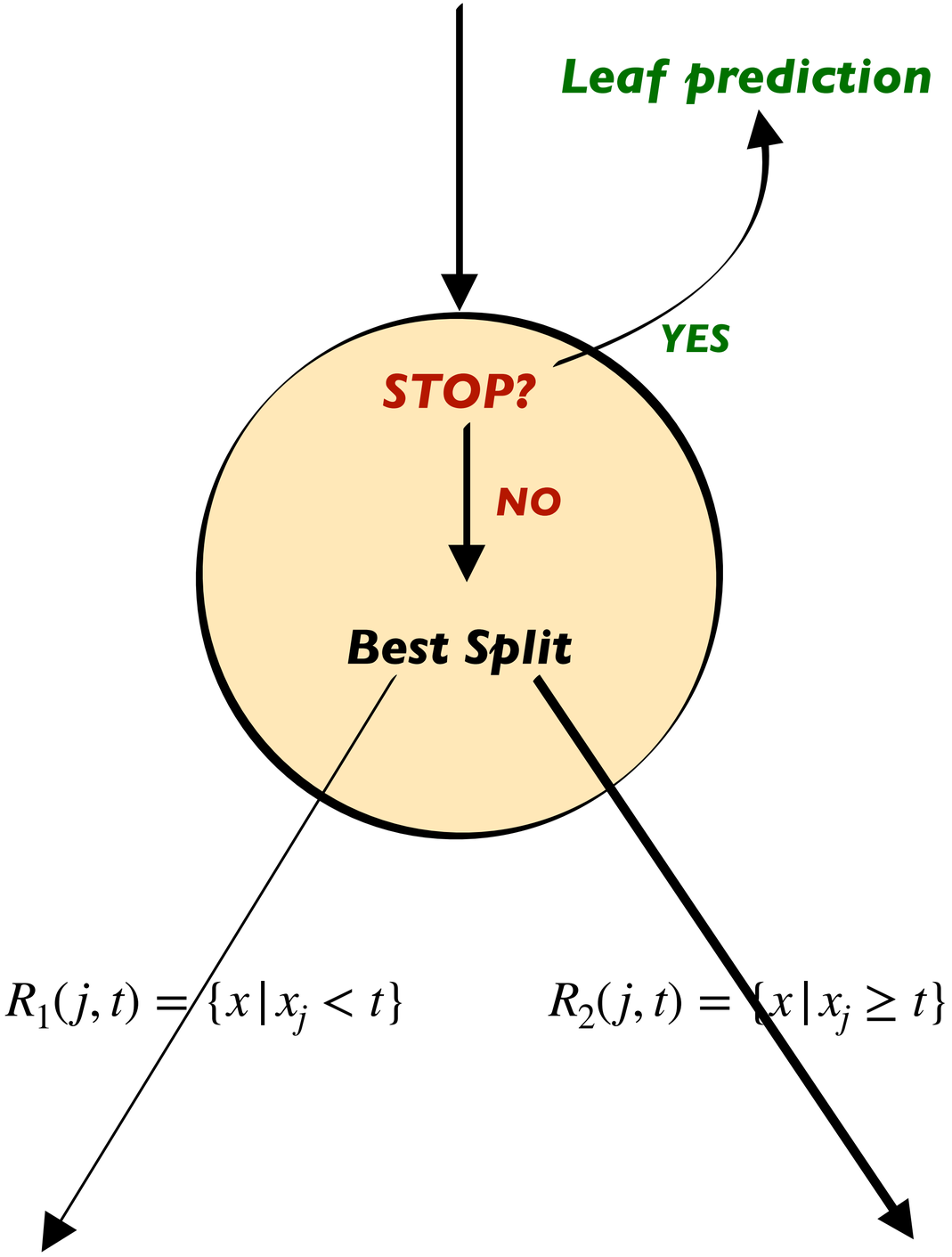
$$j, t = \text{argmax } Q(R_m, j, t) \text{ (Quality)}$$

$$R_l = \{(x, y) \in R_m \,|\, [x_j < t]\}$$

$$R_r = \{(x, y) \in R_m \,|\, [x_j \geq t]\}$$

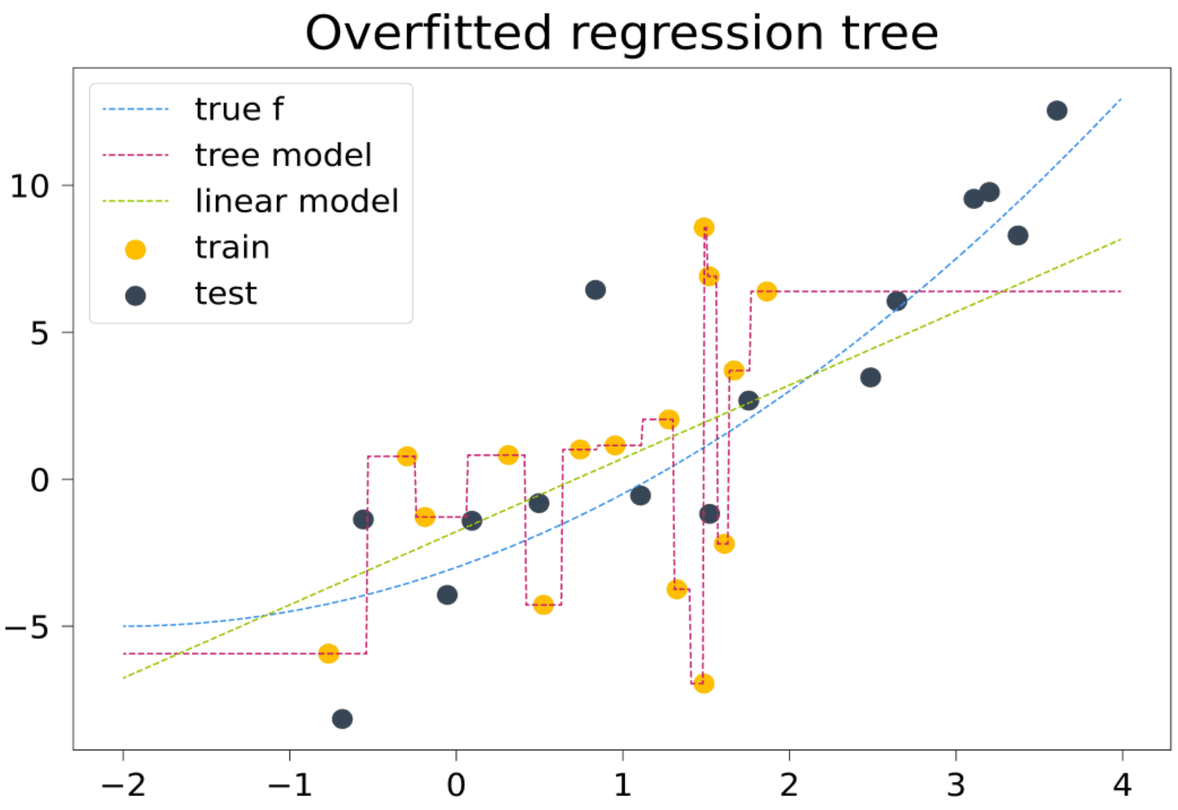
FitNode(R_l, l)

FitNode(R_r, r)

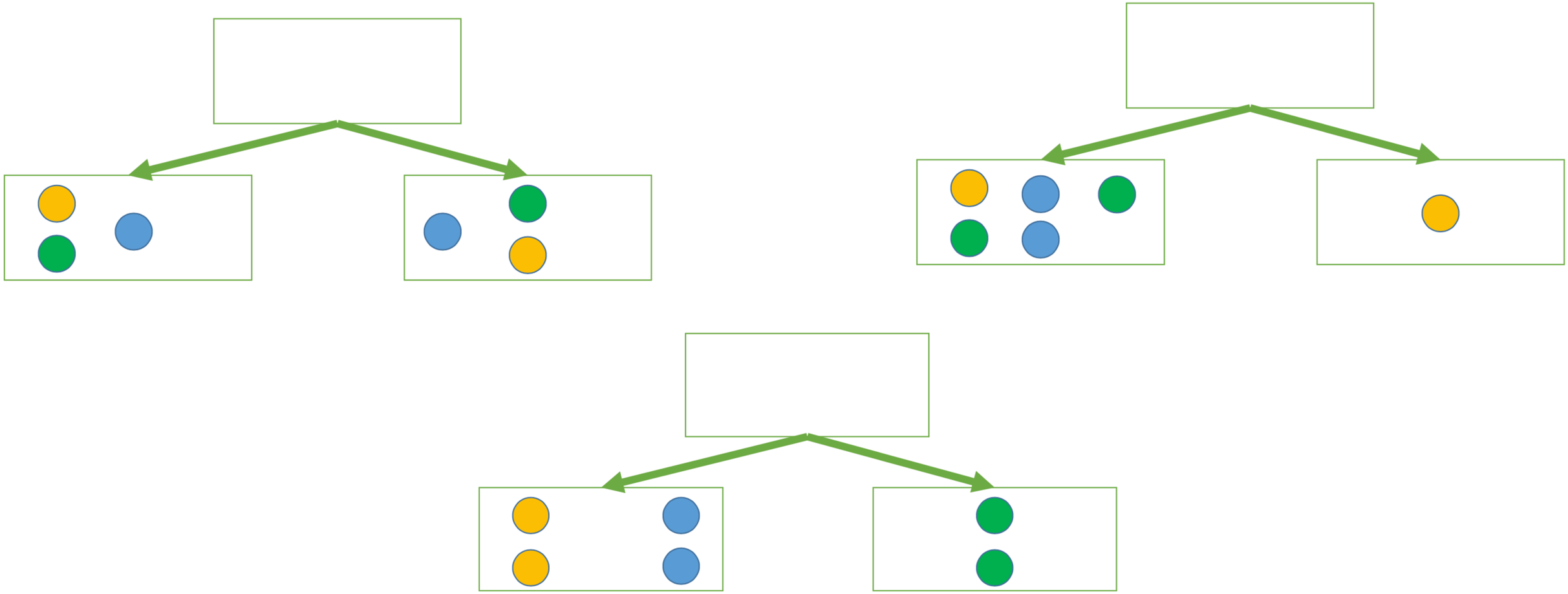


5. **Stop** criteria

- *According to **Maximum Depth***
- *If* $|R_m| < k$, *so № of items in node is below some constant*
- *If error of constant prediction for R_m is below some specific level*



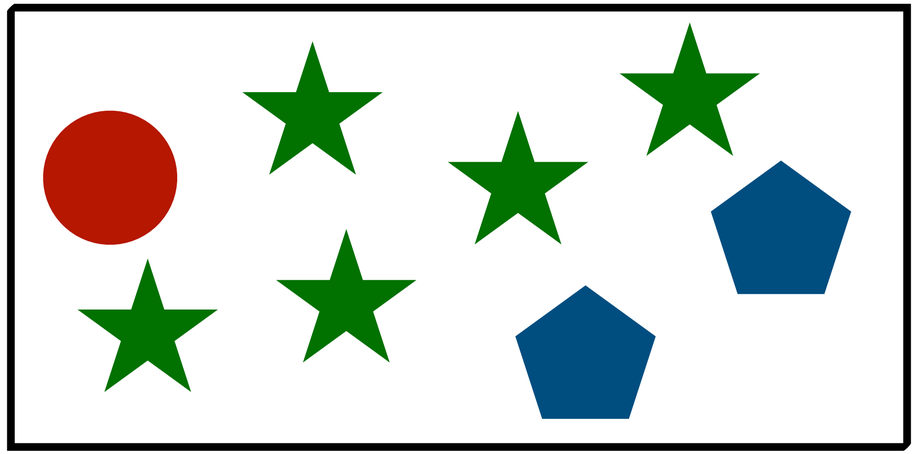
6. **Impurity**/лучшее разбиение?



6. Impurity

- $H(R_m)$ - *impurity (критерий информативности)*
  - $H(R_m) \uparrow \iff y \in R_m$ *разнообразный*

- *Общий вид:* $H(R_m) = \min\limits_{c \in \mathbb{Y}} \dfrac{1}{|R|} \sum\limits_{(x,y) \in R} L(y, c)$



$L(y, c) = [y \neq c]$

$c =$

$H(R) = \dfrac{3}{8}$

6. Impurity

- $H(R_m)$ - *impurity (критерий информативности)*
  - $H(R_m) \uparrow \iff y \in R_m$ *разнообразный*

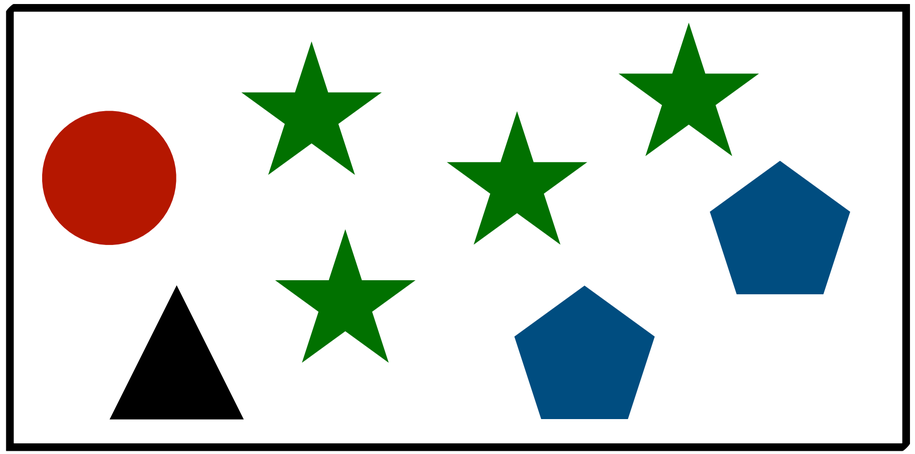
- ***Регрессия***: $$H(R) = \min_{c \in \mathbb{Y}} \frac{1}{|R|} \sum_{(x,y) \in R} (y_i - c)^2 = \frac{1}{|R|} \sum_{(x,y) \in R} (y_i - \hat{y}_R)^2$$

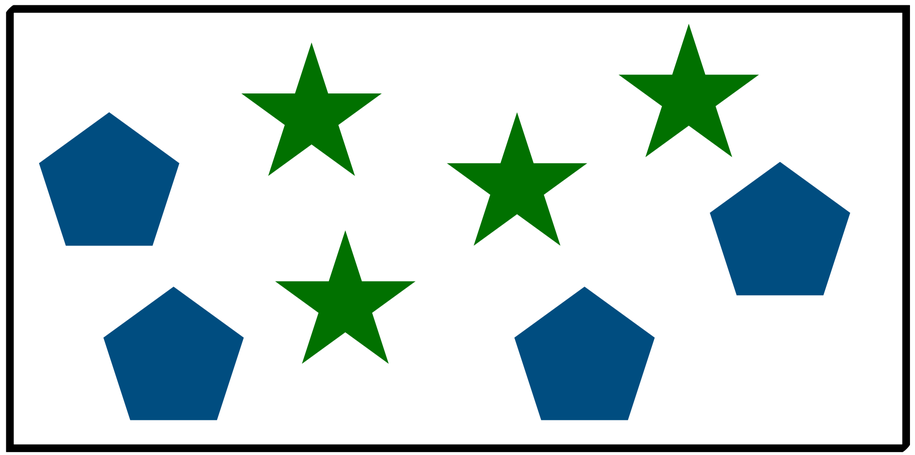
  - *Где* $\hat{y}_R$ *— среднее значение в вершине* R
  - *Формула - дисперсия значений в вершине*
  - *А если MAE?*

6. Impurity

- $H(R_m)$ - *impurity (критерий информативности)*
  - $H(R_m) \uparrow \iff y \in R_m$ *разнообразный*

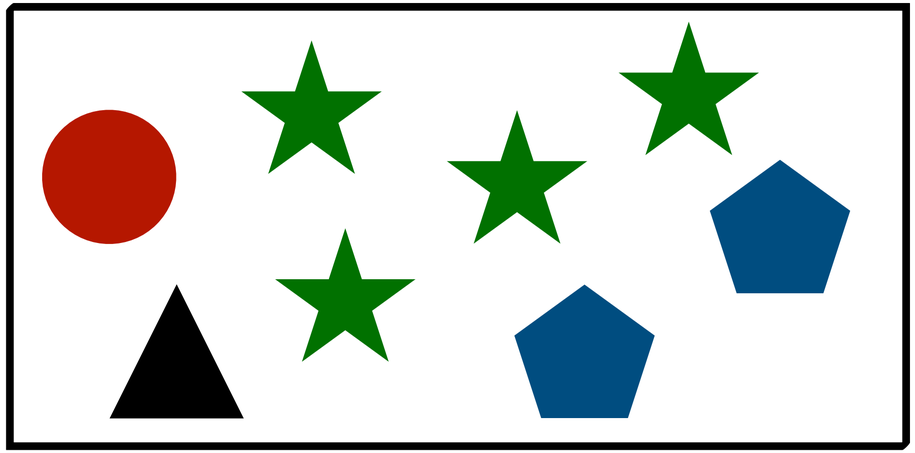
- ***Классификация***:
  - *Пусть* k *— самый частый класс в вершине*
  - *Тогда* $p_k = \dfrac{1}{|R|} \sum\limits_{(x,y)\in R} [y = k]$
  - $H(R) = 1 - p_k$



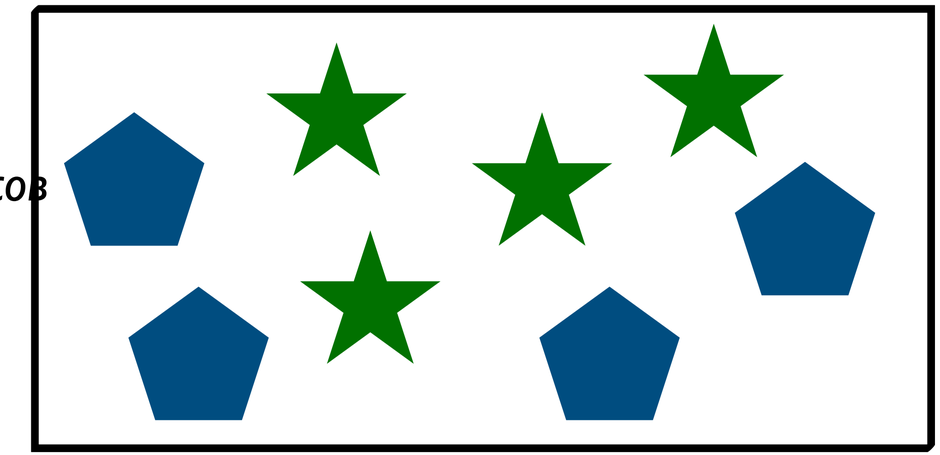


6. Impurity

- $H(R_m)$ - *impurity (критерий информативности)*
  - $H(R_m) \uparrow \iff y \in R_m$ *разнообразный*



- *Классификация:* ***Gini criteria***
  - $$H(R) = \min_{\sum c_k = 1} \frac{1}{|R|} \sum_{(x_i, y_i) \in R} \sum_{k=1}^{K} (c_k - [y_i = k])^2$$
  - *Здесь* K *— кол-во классов*
  - $c = (c_1, \ldots c_K)$ *— распределение вероятностей классов*
  - *Оптимальный* $c = (p_1, \ldots p_K)$
  - *Тогда итоговый Критерий:* $H(R) = \sum^{K} p_k(1 - p_k)$



6. Impurity

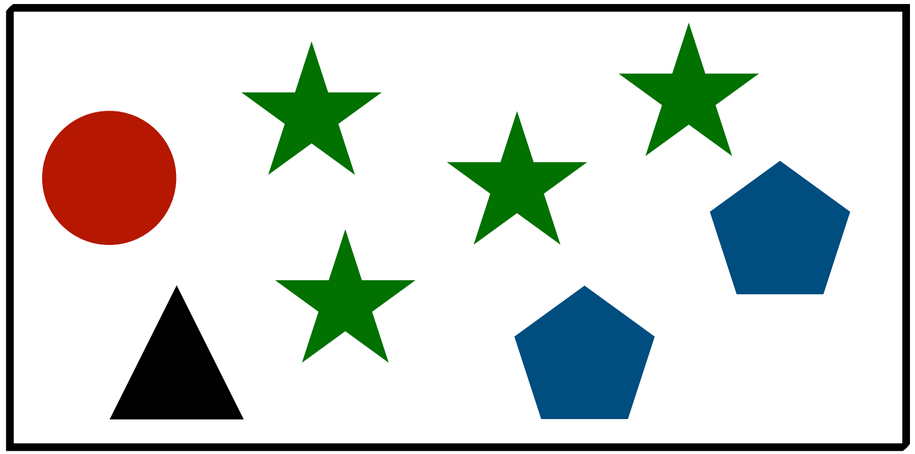
$$H(R) = \frac{1}{|R|} \sum_{(x_i,y_i)\in R} \sum_{k=1}^{K} (p_k - [y_i = k])^2$$

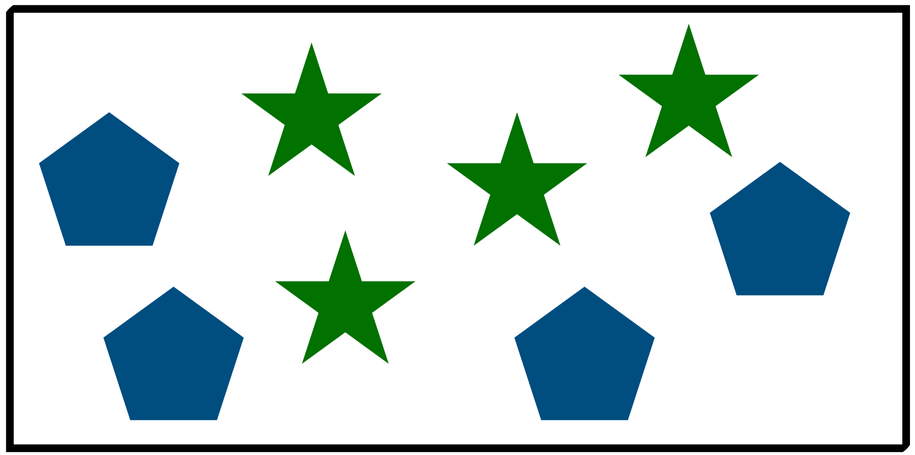
$$H(R) = \sum_{k=1}^{K} \frac{1}{|R|} \sum_{(x_i,y_i)\in R} p_k^2 + [y_i = k]^2 - 2p_k[y_i = k]$$

$$p_k = \frac{\sum_{(x_i,y_i)\in R} [y_i = k]}{|R|} \rightarrow \sum_{(x_i,y_i)\in R} [y_i = k] = p_k \cdot |R|$$

$$H(R) = \sum_{k=1}^{K} \frac{1}{|R|} (|R| \cdot p_k^2 + p_k|R| - 2p_k|R|p_k)$$

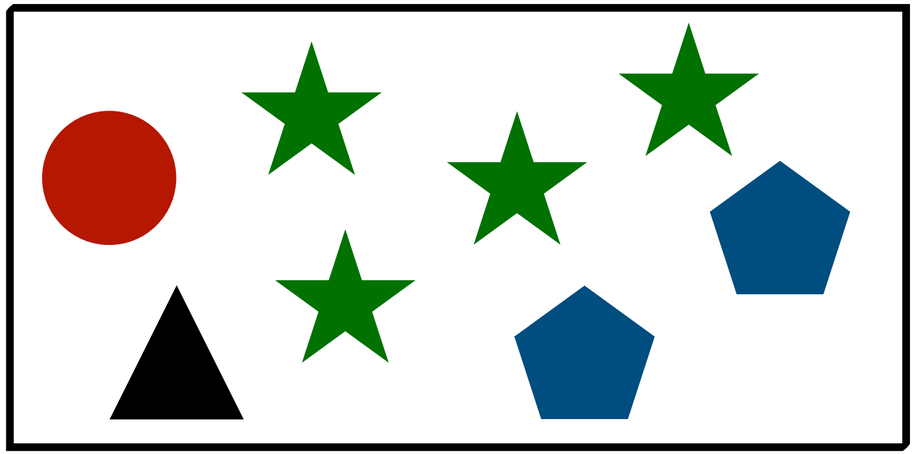
$$H(R) = \sum_{k=1}^{K} p_k^2 + p_k - 2p_k^2 = \min \sum_{k=1}^{K} p_k(1 - p_k)$$





6. Impurity

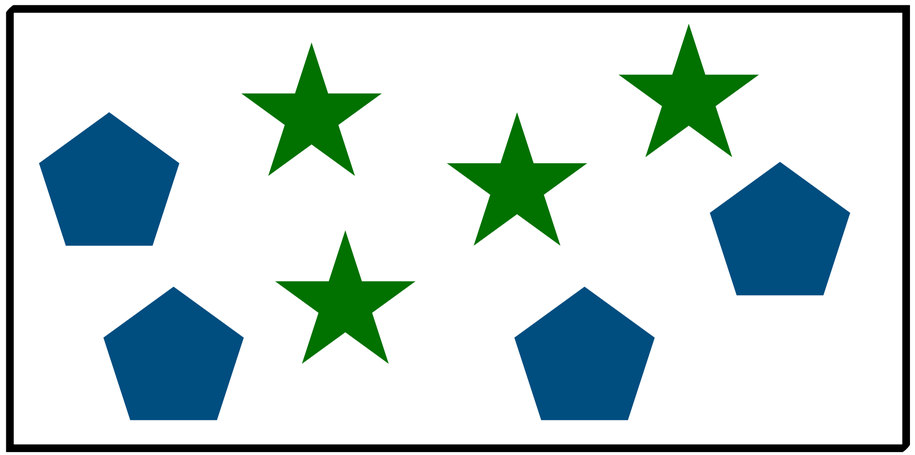
- $H(R_m)$ - *impurity (критерий информативности)*
  - $H(R_m) \uparrow \iff y \in R_m$ *разнообразный*



- *Классификация:* ***Entropy criteria***
  - $$H(R) = \min_{\sum c_k = 1} \left(-\frac{1}{|R|} \sum_{(x_i, y_i) \in R} \sum^{K} \log(c_k)[y_i = k]\right)$$
  - *Оптимизирует данный лосс, получаем:*

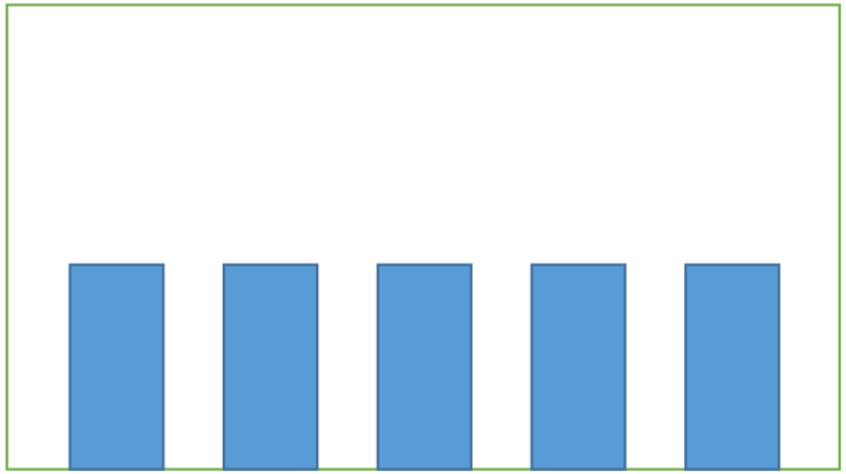
$$H(R) = -\sum^{K} p_k \log(p_k)$$

    - *0 если 1 класс, max. если равномерно*

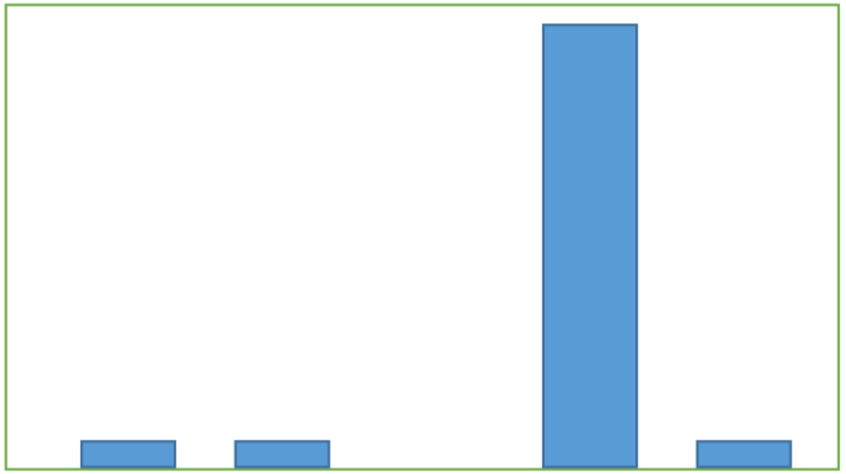


6. **Impurity**/entropy

- Мера неопределённости распределения



Высокая энтропия



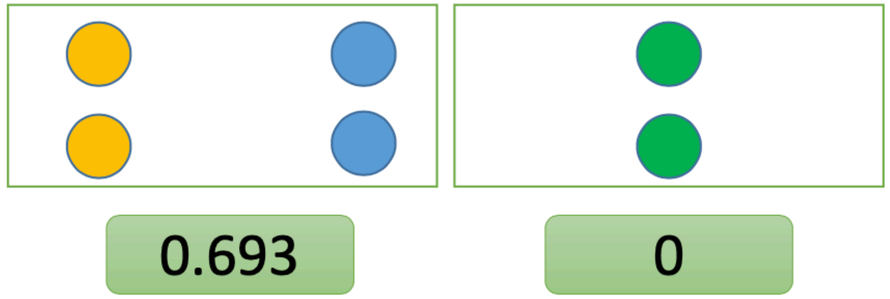
Низкая энтропия

6. **Impurity**/entropy

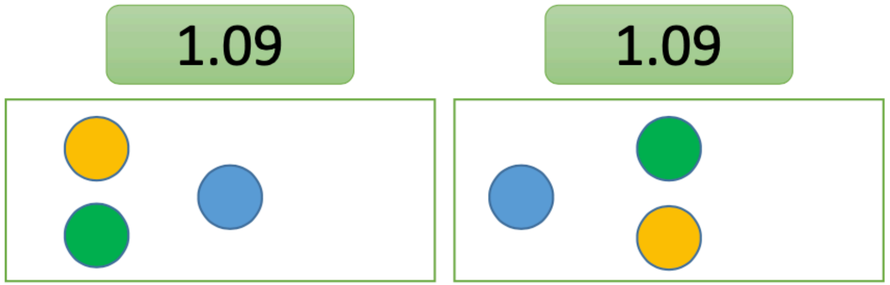
- $(0.2, 0.2, 0.2, 0.2, 0.2)$
- $H = 1.60944 \ldots$

- $(0.9, 0.05, 0.05, 0, 0)$
- $H = 0.394398 \ldots$

- $(0, 0, 0, 1, 0)$
- $H = 0$



- $(0.5, 0.5, 0)$ и $(0, 0, 1)$
- $H = 0.693 + 0 = 0.693$



- $(0.33, 0.33, 0.33)$ и $(0.33, 0.33, 0.33)$
- $H = 1.09 + 1.09 = 2.18$

6. Impurity: **Итог**

- Как понять, какой предикат лучше?
- Сравнить хаотичность в исходной вершине и в двух дочерних!

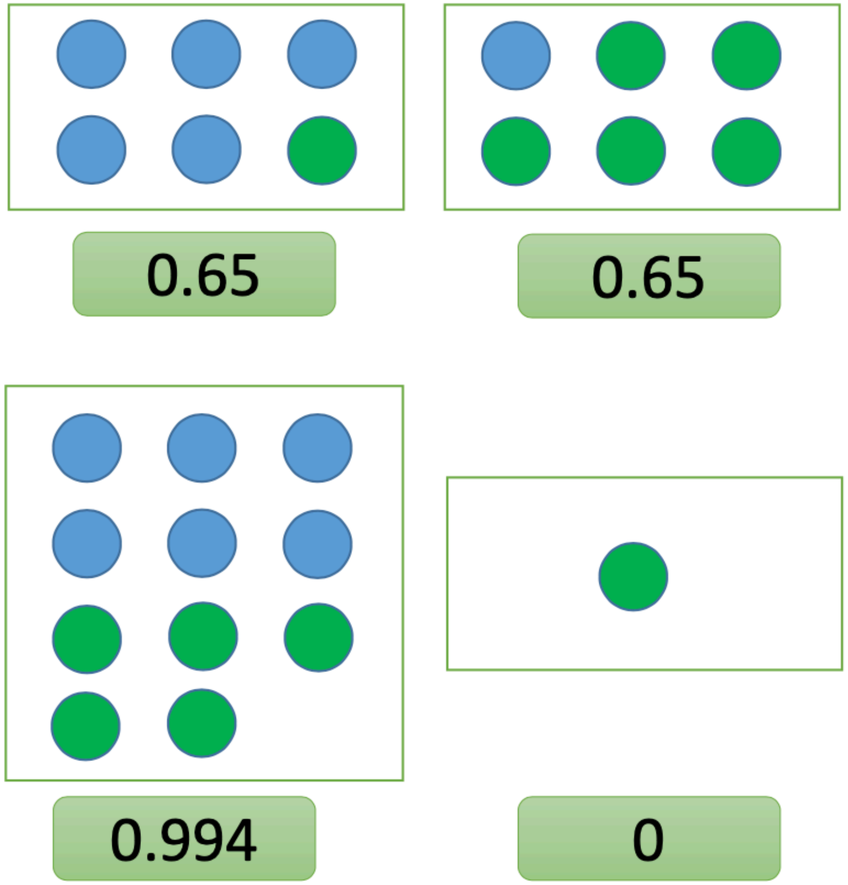
6. Impurity: **Итог**

- $H(R_m)$ - *impurity (критерий информативности)*
  - $H(R_m) \uparrow \iff y \in R_m$ *разнообразный*

- $$Q(R_m, j, t) = H(R_m) - \frac{|R_l|}{|R_m|} H(R_l) - \frac{|R_r|}{|R_m|} H(R_r) \longrightarrow \max$$

Доля объектов дочерней вершины
~ вес ее impurity

6. Impurity: **Итог**



- $(5/6, 1/6)$ и $(1/6, 5/6)$
- $0.65 + 0.65 = 1.3$

- $(6/11, 5/11)$ и $(0, 1)$
- $0.994 + 0 = 0.994$

7. Miscellaneous: empty / categorical / pruning

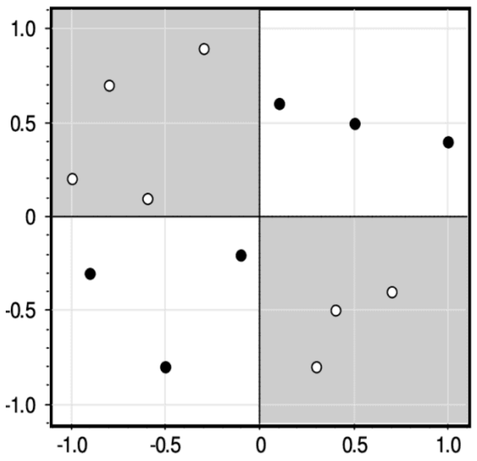
- *Pruning - стрижка деревьев*
  - *Строим переобученное дерево - оптимизируем его структуру*
  - *Намного лучше раннего останова*
  - *Но! Деревья никому не интересны :(*
    - *E.G. Cost-Complexity pruning*

- *Empty values at feature* j
  - *Training:* $Q(R, j, t) = \frac{R \text{without } V_j}{|R|} Q(R \text{ without } V_j, j, t)$
  - *Inference:* $a_{mk}(x) = \frac{|R_l|}{|R_m|} a_{lk}(x) + \frac{|R_r|}{|R_m|} a_{rk}(x)$
  - *You can also use surrogate predicates!*

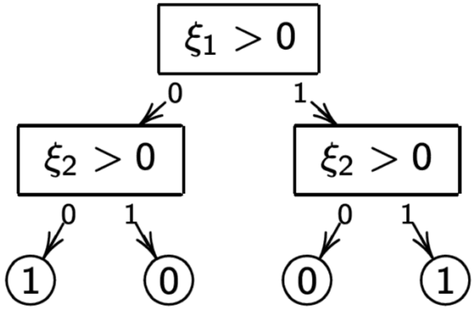
7. Miscellaneous: empty / categorical / pruning

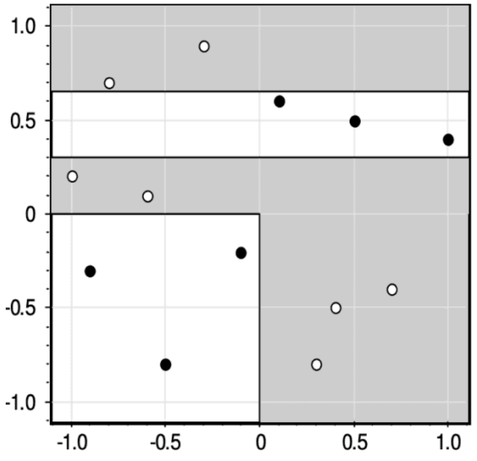
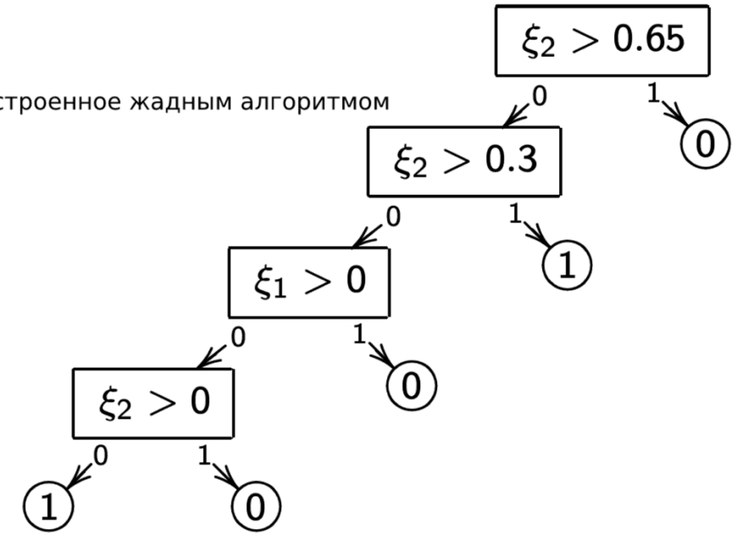
- *Categorical values*
  - *Multi-way splits — too many leaves*
  - $U = (u_1, \ldots, u_q) = U_1 \cup U_2$ - *множество значений j-ого признака*
    - *Too many options!*
  - $R(u_1)$ — *all objects with value* u_1 *at feature j*
    - $\frac{1}{N_{u_1}} \sum_{x_i \in R_{u_1}} [y_i = +1]$ — *доля положительных*
    - $\frac{1}{N_{u_{(1)}}} \sum_{x_i \in R_{u_{(1)}}} [y_i = +1] \leq \ldots \leq \frac{1}{N_{u_{(q)}}} \sum_{x_i \in R_{u_{(q)}}} [y_i = +1]$
    - $(1) \to 1; (2) \to 2; \ldots; (q) \to q \Longrightarrow$ *числовой признак*

Well…



Оптимальное дерево





Дерево,построенное жадным алгоритмом